QUERY CARTEL · CLICK-A-THON 2026 · ATLYS TRACK

Analytics Copilot

Feature spec in. Production schema + PM-ready insights out.

Agents instrument, analyze, and explain — fully traced.

Collapse weeks of instrumentation handoffs into one chat:
schema designed, context kept fresh, insights with charts & tables — every step traced.

ClickHouse · Deterministic agents · LLM as the brain · Langfuse proof

THE CHALLENGE

Feature shipping outpaces analytics

Manual loop

Tracking PRD → schema → analysis takes weeks. Context dies in handoffs.

Stale insight

By the time numbers land, the team has moved on. No why, only what.

Unseen-spec day

Day-2 sealed feature must run through the same pipeline — no hardcoding.

Goal: collapse the loop — instrument, analyze, explain — on ClickHouse

1 Instrumentation

Spec → production DDL (ORDER BY, partition, TTL, MVs)

2 Analytics

SQL-backed insights a PM would act on

3 Context

Living business layer; fresh after every table

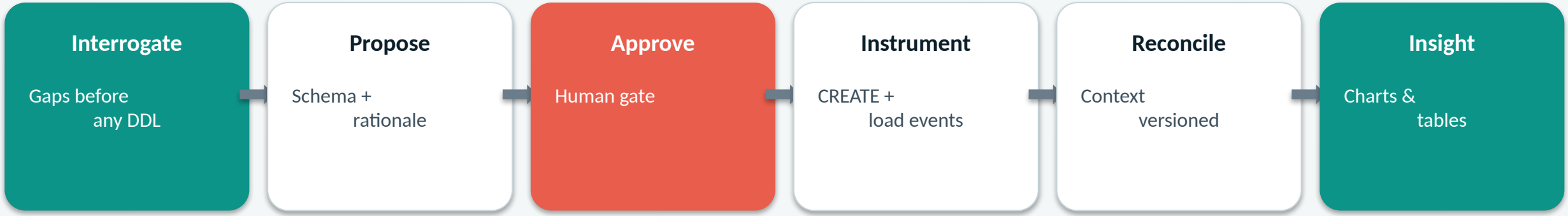
4 Trace + UI

Langfuse proof + charts, tables, diffs

OUR SOLUTION

Feature spec in. Schema + insight out.

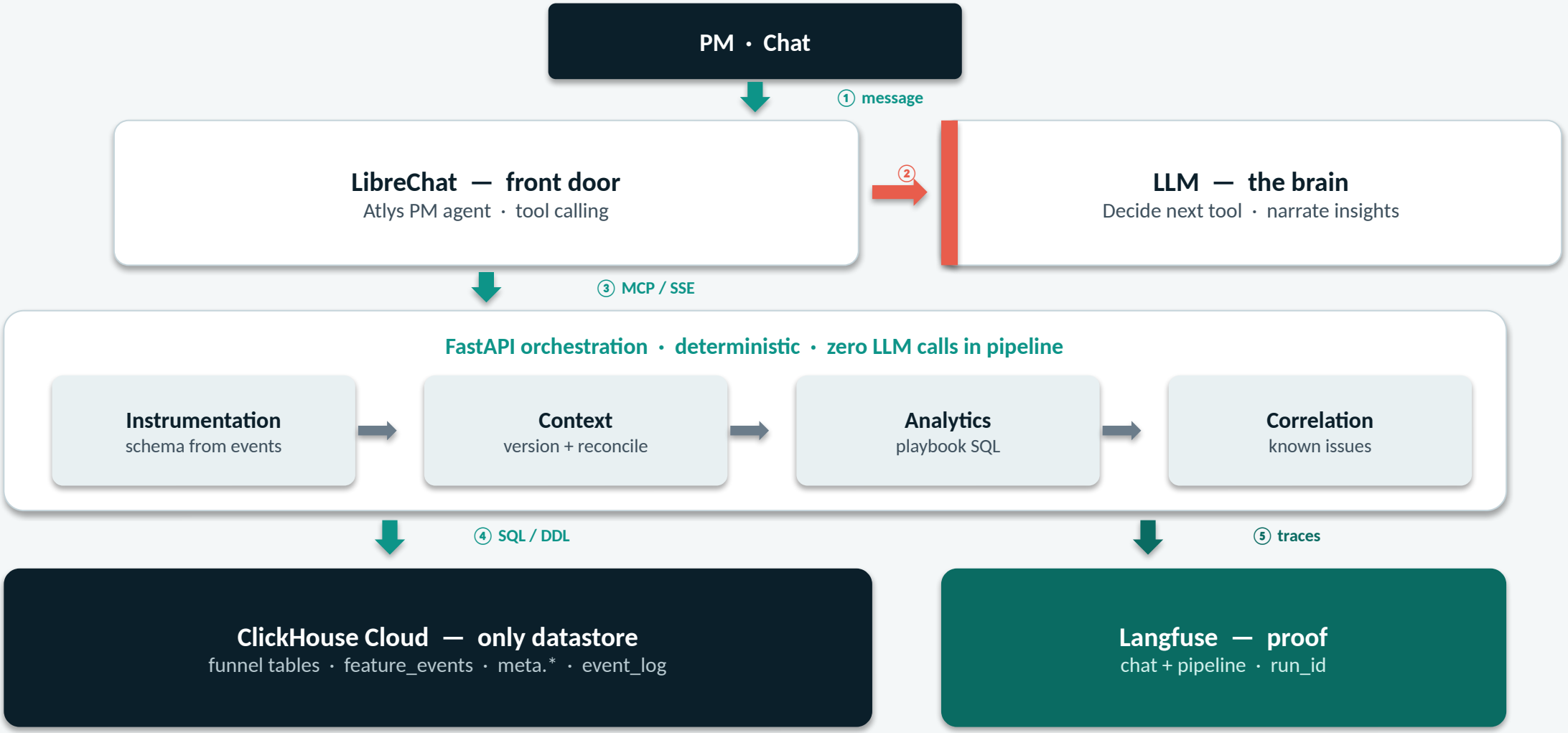
Agents instrument, analyze, and explain — fully traced. Charts & tables included.



Deterministic-first	Pipeline owns schema & numbers with zero LLM calls. The LLM is the brain for chat & narration — ClickHouse owns the math.
Compute in ClickHouse	Playbook SQL pushes aggregates into CH. The brain never sees raw rows — protects token budget & accuracy.
Generic for Day 2	Content-driven pipeline. No feature names hardcoded. Same chat flow for the sealed sixth spec.

SYSTEM DESIGN

Architecture

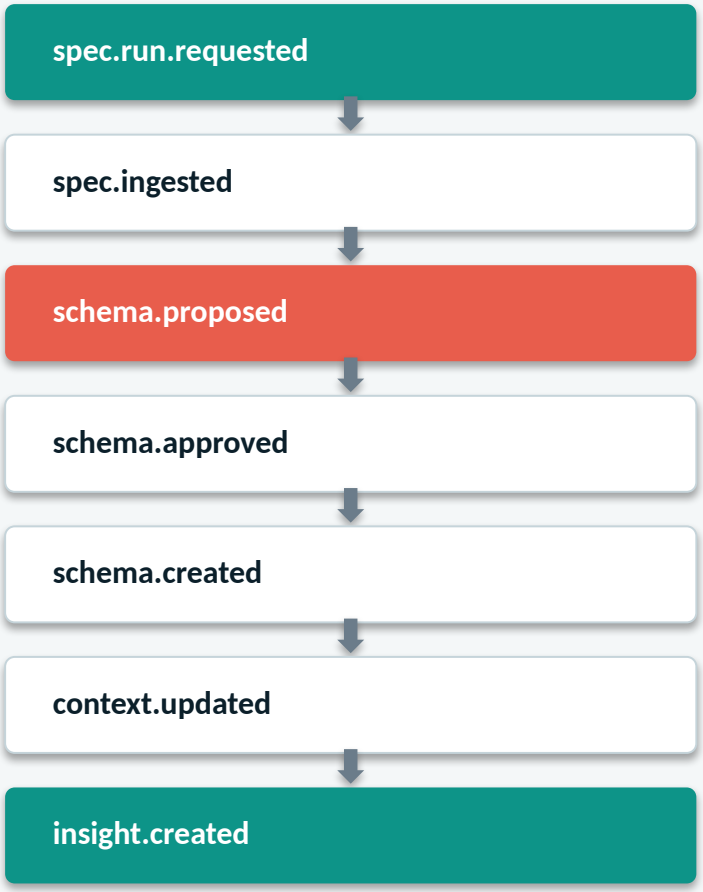


⑧ Dashboard REST ← FastAPI · React shell shows insights, charts & tables, schema timeline, context diffs

ORCHESTRATION

Event-driven pipeline

Agents are event handlers. atlys.event_log is the durable backbone.



Why this design

Approval gate

Pauses at schema.proposed. Pending run in meta.pending_runs. DDL only after approve_schema.

Idempotent & auditable

Every transition → event_log row + Langfuse span. Replay the full chain from one trace_id.

Context before insight

Analytics never runs on a stale snapshot. context.updated always precedes insight.created.

Chat is a thin client

The LLM brain picks tools; FastAPI owns state. Reloads don't lose a pending run.

DATA

ClickHouse data model

EXISTING FUNNEL (~2.5M rows)

1. **destination_card_clicked**
2. **application_started**
3. **document_uploaded**
4. **purchase_completed**

+ supporting: search · scroll · auth · pay_now

Per feature: <feature>_events
Designed from the spec's NDJSON — types, ORDER BY, partition, TTL — after approval.

PIPELINE META (auto-created)

schema_catalog / changelog	DDL, rationale, versions
context_snapshots	versioned business layer
context_changelog	diffs + contradictions
insights	cards + evidence + SQL
pending_runs	approval gate state
known_issues	linked from context
migration_journal	safe DDL apply
atlys.event_log	durable event backbone

WHY WE WIN

Mapped to judging criteria

01	**Schema quality**	Event-driven typing from sample NDJSON. Defensible ORDER BY / partition / TTL. MVs that earn their keep. Human approval before execute.
02	**Insight quality**	Playbook cuts across device, geo, funnel stage. Known-issue correlation. Charts & tables backed by stored SQL — PM prose, not DB dumps.
03	**Context freshness**	Context Agent versions snapshots + diffs on every schema change. Analytics always reads the latest snapshot.
04	**Traceability**	Langfuse dual-trace (chat + pipeline), cross-linked by session_id = run_id. meta.* and event_log carry trace_id.
05	**Unseen spec**	Zero per-spec hardcoding. Same interrogate → propose → approve → analyze path. Built for the sealed sixth spec.

QUERY CARTEL · LIVE DEMO

Feature spec in. Production schema + PM-ready insights out.

Agents instrument, analyze, and explain — fully traced.

1 Upload / pick a feature spec → "implement Express Checkout"

2 Agent interrogates gaps → proposes DDL + rationale

3 Approve in chat → CREATE TABLE + load + context reconcile

4 Insight card + charts & tables + evidence SQL + Langfuse link

5 Dashboard: changelog · context diff · event_log replay

Stack

ClickHouse Cloud · FastAPI / MCP · LibreChat · LLM (brain) · Langfuse · React